Field Report:
Surface Water–Groundwater Investigation

Date of Fieldwork: October 31, 2025

Time: 8:00 AM – 2:00 PM

Location: UF–IFAS Southwest Florida Research and Education Center (SWFREC), Immokalee, FL

Host: Dr. Nikolaos Tziolas, Soil Science AI Lab, University of Florida (UF)

Team: attendees from FGCU and UF students

Purpose and attendees

The field visit aimed to integrate hands-on investigation of surface water–groundwater, downloading data from weather station and soil moisture sensors, field permeameter, borehole drilling and head measurement, electrical resistivity tomography.

FGCU Attendees

- 2 TWS undergraduate students
- 3 WCE undergraduate students
- 4 WCE graduate students
- 1 WCE postdoc
- 2 TWS and WCE faculty

UF Attendees

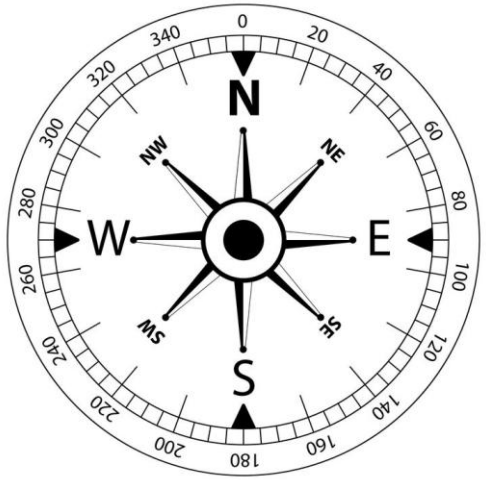
- 4 PhD students
- 1 faculty

Fieldwork Activities

Time	Activity	Description
08:00 – 08:30 AM	Arrival and Introduction	Orientation at SWFREC reception, overview of field objectives
08:30 – 12:30 PM	Field Data Collection	• Collected meteorological and soil moisture data (15-min intervals). • Drilled three shallow boreholes (BH1–BH3) and measured hydraulic head levels. • Conducted a 24-electrode ERT resistivity survey along the designated transect. • Field permeameter (display only, demo failed due to charging)
12:30 – 01:00 PM	Lunch Break	Group lunch at SWFREC auditorium.
01:00 – 02:00 PM	Lab Visit	Tour of the Soil Science AI Lab (robots, UAV hyperspectral systems, digital-twin demonstrations).

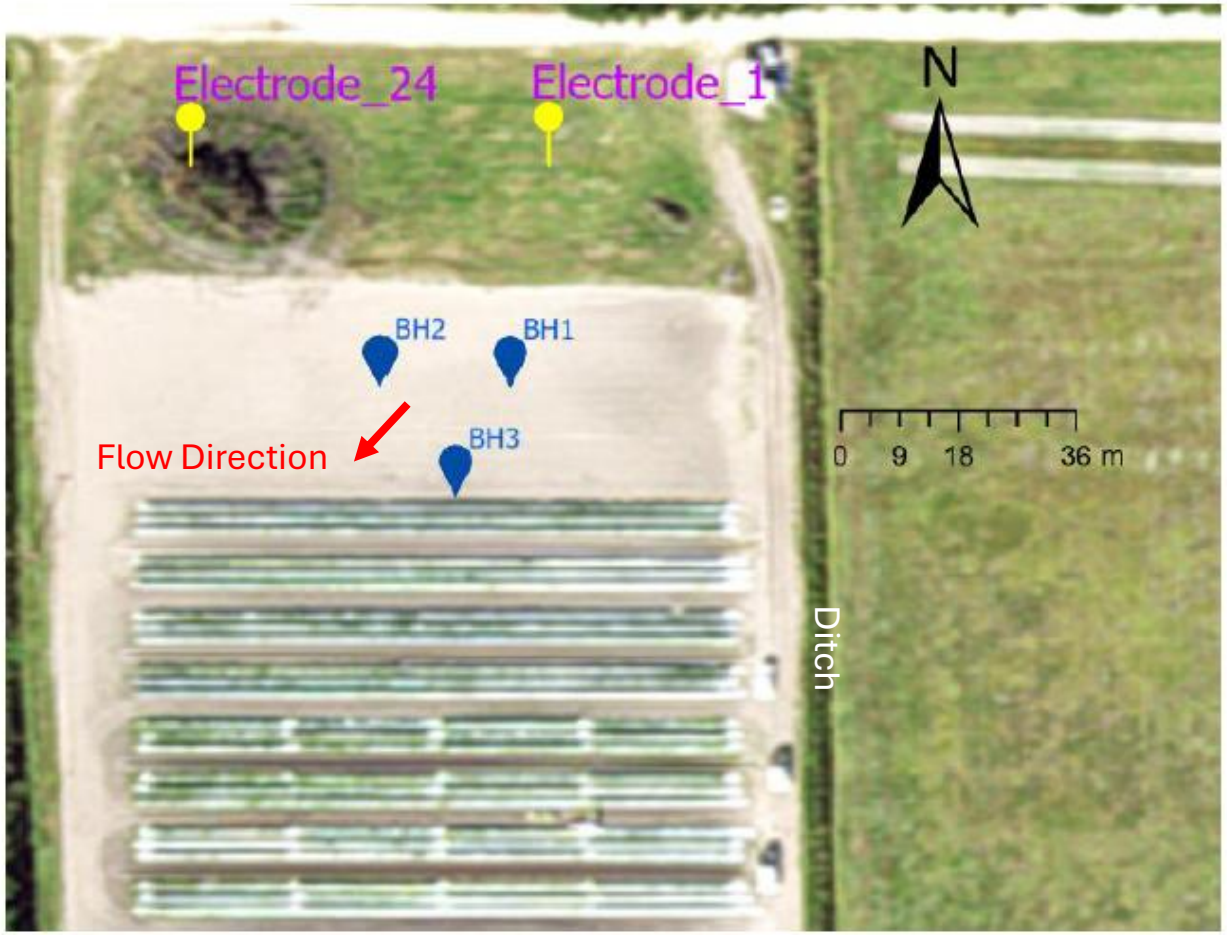
Boreholes Drilled

ID	Latitude	Longitude	Depth (ft)
BH1	26.4621	-81.4423	2.90
BH2	26.4621	-81.4425	3.15
BH3	26.46195	-81.4424	3.25



Flow Direction: South-West
Flow Azimuth: 222.9° (measured clockwise from North)
Hydraulic Gradient i: 0.0184 ft/m

High Head: BH1(9.97 m, 16.67 m, - 2.90 ft)
Medium Head: BH2 (-9.97 m, 16.67 m, -3.15 ft)
Low Head: BH3 (0,0, -3.25 ft)



This means the water table drops 0.0184 feet for every 1 meter of horizontal distance in the flow direction

Boreholes Drilled

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BH1	26.4621	-81.4423	2.90
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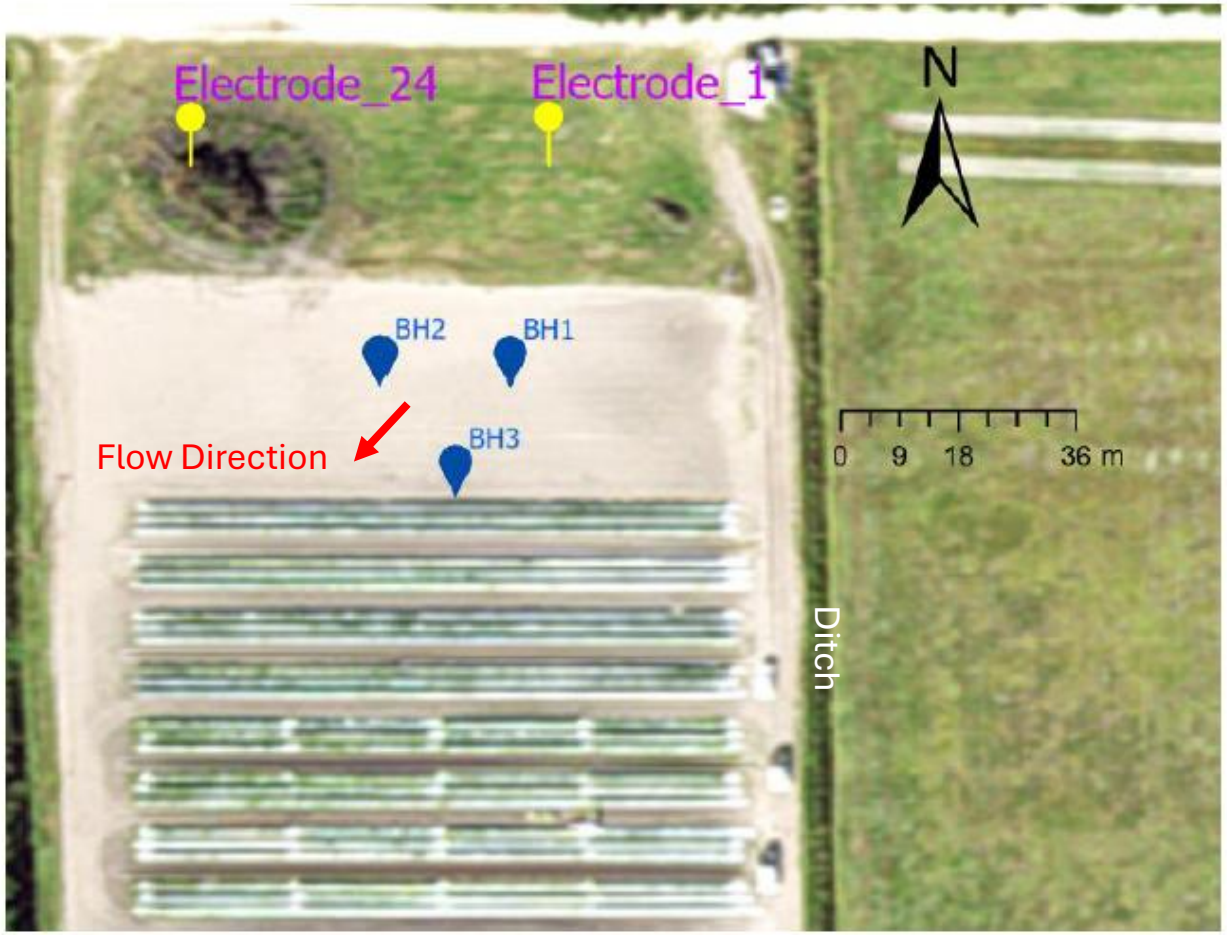
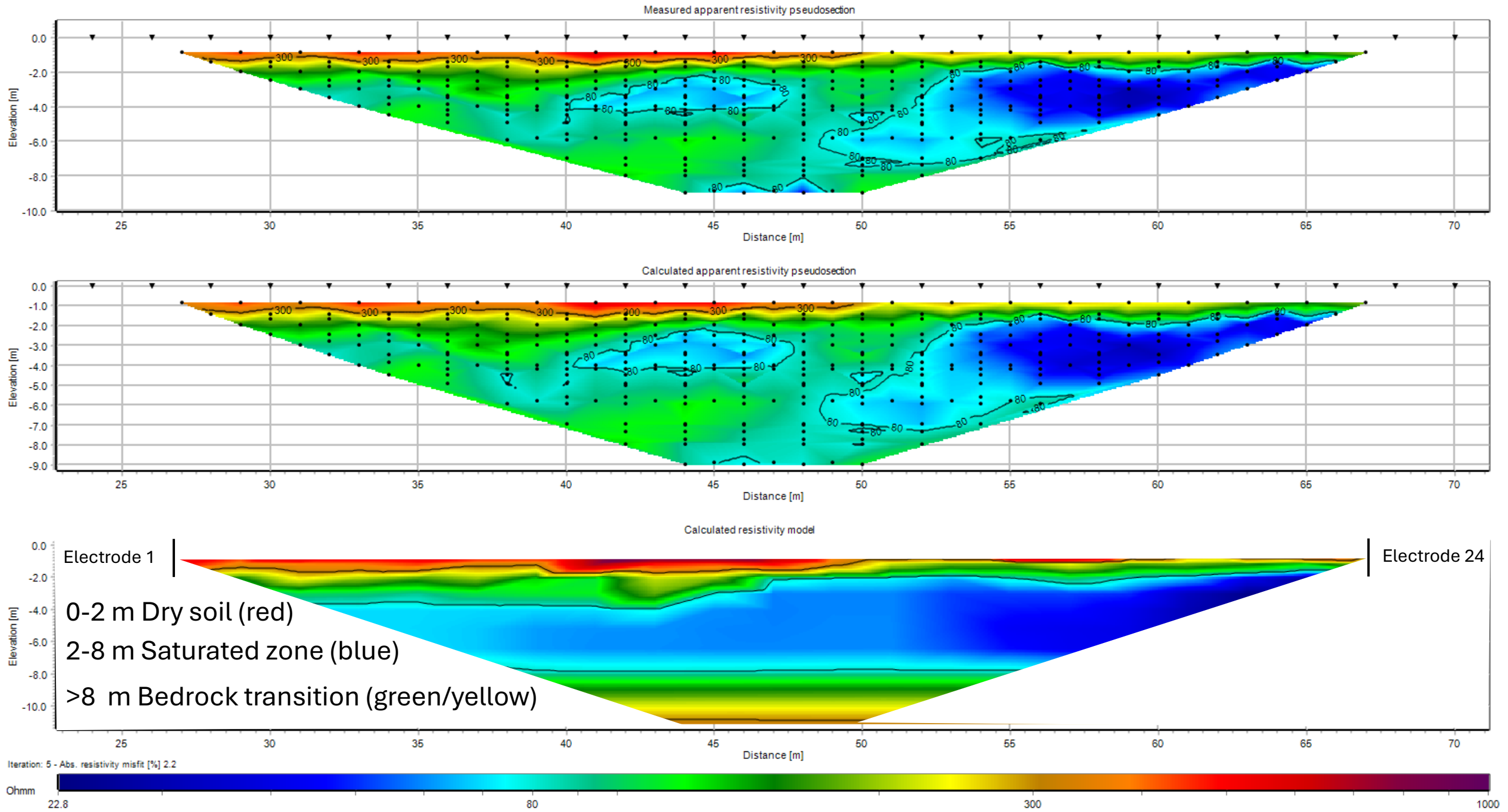
BH3	26.46195	-81.4424	3.25

ERT Line

Electrode	Latitude	Longitude
E1	26.4624	-81.4423
E24	26.4624	-81.4428

High Head: BH1(9.97 m, 16.67 m, - 2.90 ft)
Medium Head: BH2 (-9.97 m, 16.67 m, -3.15 ft)
Low Head: BH3 (0,0, -3.25 ft)




Measured apparent resistivity pseudosection
Calculated apparent resistivity pseudosection
Calculated resistivity model
Elevation [m]
Distance [m]
Electrode 1
Electrode 24
0-2 m Dry soil (red)
2-8 m Saturated zone (blue)
>8 m Bedrock transition (green/yellow)
Iteration: 5 - Abs. resistivity misfit [%] 2.2
Ohmm
22.8
80
300
1000

Meteorological and soil moisture data

All atmospheric variables from the ATMO 41 station and soil moisture sensors were recorded every 15 minutes through PORT 1 to PORT 5.

Port	Sensor	Depth(s)/height	Measured Parameters
1	ATMOS 41	2 m above ground	Solar radiation, air temperature, humidity, pressure, wind speed, guest speed, precipitation, lightning activity
2	TEROS 12	15 cm	Volumetric water content, soil temperature, electrical conductivity (EC)
3	TEROS 12	30 cm	Volumetric water content, soil temperature, EC
4	TEROS 12	45 cm	Volumetric water content, soil temperature, EC
5	TEROS 54	15, 30, 45, 60 cm	Volumetric water content and soil temperature at four depths

- The ATMO 41 recorded continuous data with minor intermittent NaN values in the wind, maybe due to transient signal quality.
- Soil moisture sensors provided stable readings across depths

Borehole drilling for head measurement

ERT transect

Field permeameter

AI lab of Dr. Nicos Tziolas

Conclusions

- The field campaign at SWFREC successfully achieved its objectives except for hydraulic conductivity measurements
- Three boreholes were drilled, a 24-electrode ERT line was established, and all meteorological parameters and soil moisture data at 15-minute resolution were successfully retrieved for analysis.
- This integrated dataset will support future modeling of soil–water–atmosphere interactions.
- Fieldwork and coordination with Dr. Tziolas's lab were efficient, enhancing practical exposure to advanced sensing technologies.